

Methodology

Integrated HC algorithm into widely used pipelines and constructed Null Distribution over several popular datasets which cover many different topics.

Pipelines used

- Adaptive-k Retrieval (megagonlabs EMNLP 2025): Implementation of adaptive thresholding method. This method is especially effective for long-context QA tasks in RAG. Given a list with similarity scores for some query, it chooses the number of documents to be retrieved by finding the largest first gap in scores.
- Adaptive-RAG (Jeong et.al NAACL 2024): Implementation of adaptive QA router, instead of always using the same RAG strategy, it first estimates how complex the question is and then automatically chooses one of three modes: no-retrieval (NOR) for easy questions, single-step retrieval (OneR) for medium questions, and iterative multi-hop retrieval (IRCoT) for hard questions. The routing is done by a small classifier model trained on automatically collected labels.
- FlashRAG (Jin et.al WWW2025): Research and benchmarking toolkit for building and comparing RAG pipelines quickly and consistently. It provides ready building blocks: retrievers, rerankers and generation pipeline. It provides large set of preprocessed benchmark datasets and many implemented RAG methods to reproduce published baselines.
- Our pipeline: Complete Development and testing pipeline. Processing datasets with several chunking strategies and indexing using Faiss. Configurable embedding models from HuggingFace. Generation with custom prompts based on task and Open Router for LLMs. Supporting various thresholding algorithms and evaluation pipeline for testing RAG.

New dataset created

Many of the datasets we've worked with are favorable to the algorithms the pipeline supports and doesn't depict a real world environment. We decided to create new dataset which will enable us to compare HC vs other retrieval algorithms in more realistic setting:

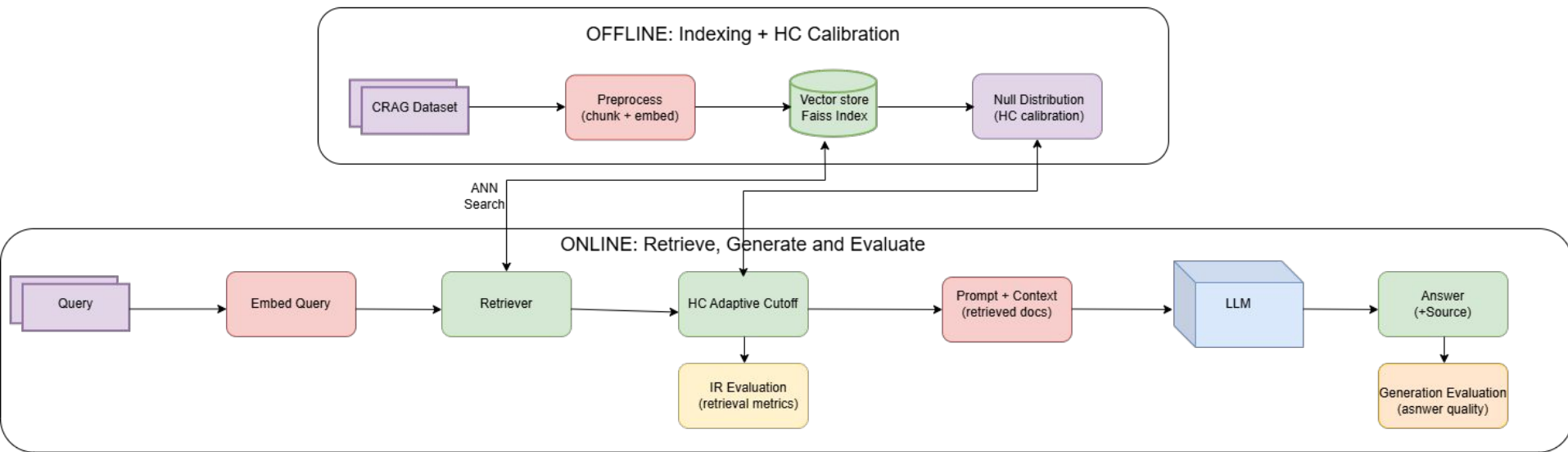
- CrossEntityQA: Dataset consists of passages on entities. Each query has a variable number of relevant entities (from 2–19 passages with avg ~8.8). Sparse corpus where only ~2% of the passages are relevant per query.
- Built from Wikipedia, including various unrelated groups of topics. LLM-generated queries and validation.

Methodology

Pipelines used - Our pipeline

Retrieval focused RAG pipeline built to evaluate Higher-Criticism (HC) adaptive retrieval on the CRAG benchmark.

HC RAG (High Level RAG Architecture)

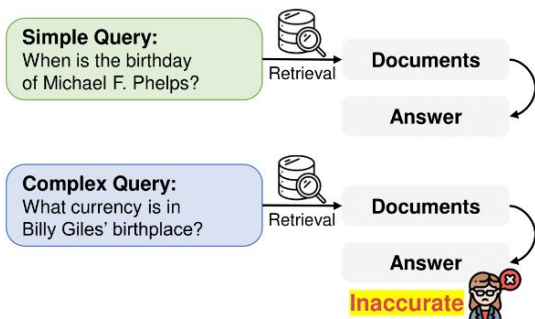


Methodology

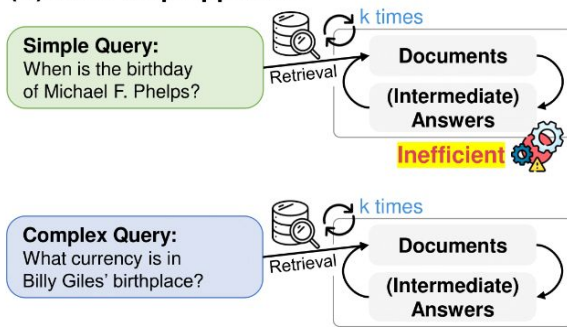
Pipelines used - Adaptive RAG (Jeong et.al NAACL 2024)

QA framework that can dynamically select the most suitable strategy for (retrieval-augmented) LLMs from the simplest to the most sophisticated ones based on the query complexity.

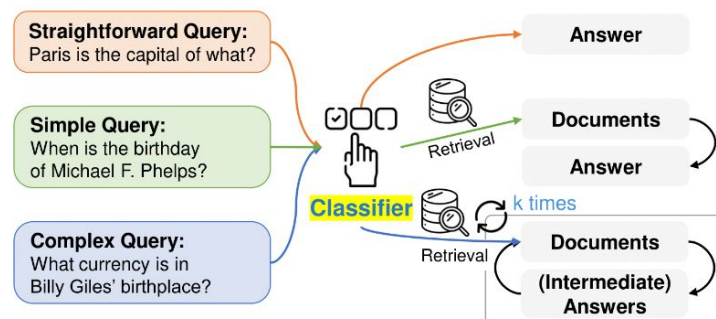
(A) Single-Step Approach



(B) Multi-Step Approach



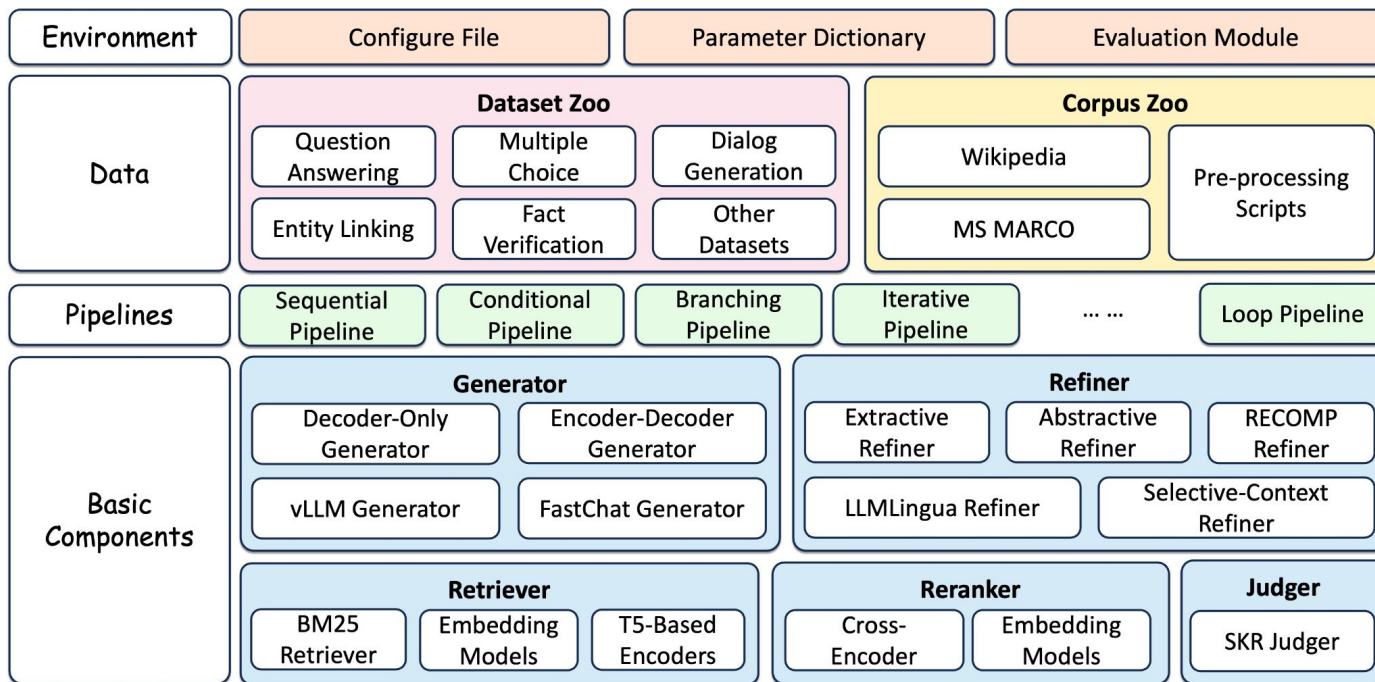
(C) Our Adaptive Approach



Methodology

Pipelines used - FlashRAG (Jin et.al WWW2025)

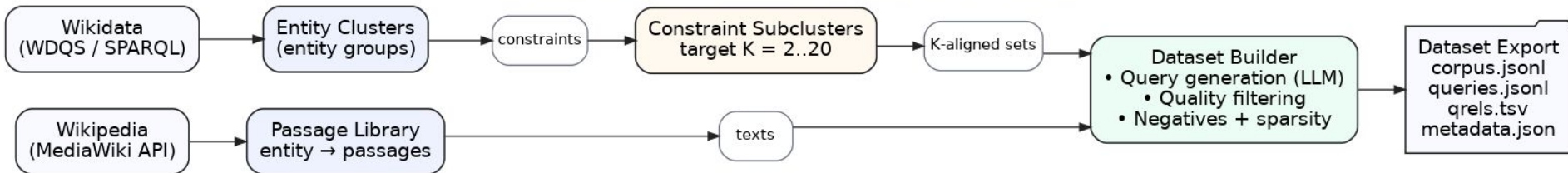
Research and benchmarking toolkit for building and comparing RAG pipelines quickly and consistently. It provides ready building blocks: retrievers, rerankers and generation pipeline.



Methodology

New dataset created - Cross Entity QA

CrossEntityQA — Dataset Construction Pipeline



- synthetic QA retrieval benchmark built from Wikipedia passages.
- Each query requires retrieving K relevant passages (with K varying from 2–20) to answer fully.
- Each passage is uniquely related to an entity across several non-related topics

Methodology

Metrics

- Retrieval: Recall@k / Precision@k + *avg context size* (avg k / token budget proxy).
- Generation: Exact Match (EM), token-level F1, Precision, Recall.

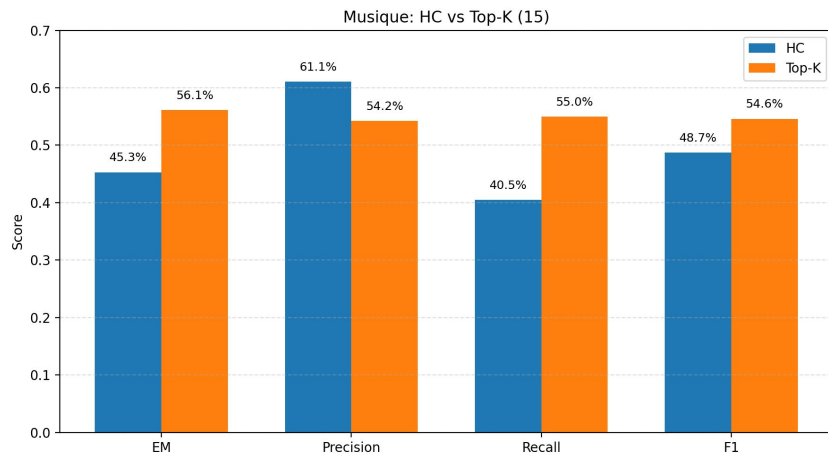
Results

- CrossEntityQA: Compared to Top-K algorithm HC achieved: **6.1%** increased precision, **50%** increased EM, **-6.5%** decreased recall
- Long context dataset (HoloBench): benchmark for testing whether long-context language models can do “database-style” reasoning over text, combining many scattered facts.
Compared to Adaptive-K algorithm HC achieved: **88%** increased precision, **41%** increased F1, **28.91%** decreased token usage and **19%** faster.
- HotpotQA: multi-hop question answering dataset built from Wikipedia, designed to test whether a system can combine evidence from multiple articles.
 - Compared to Adaptive-K algorithm HC achieved: **100%** increased precision, **60%** increased F1, **22.95%** increased EM and **94.77%** decreased token usage.
 - Compared to Adaptive RAG (multi step retrieval) HC underachieved: **-9.94%** precision, **-10.43%** F1, **-6.73%** EM.
- BEIR Scifact: Fact-checking dataset in the BEIR benchmark: each query is a short scientific claim, and the retrieval task is to find the supporting/refuting evidence in a corpus of 5,183 scientific (PubMed-style) abstracts.
Compared to Adaptive-K algorithm HC achieved: **10%** increased precision, **4%** increased recall.
- Musique: Multi-hop question answering dataset built to require genuine connected reasoning across multiple evidence pieces, rather than letting models “cheat” with a single relevant passage.
Compared to Top-K algorithm HC underachieved: **-67.9%** decreased precision, **-65%** decreased recall, **-86.4%** decreased EM.

Results - Top-k Baseline

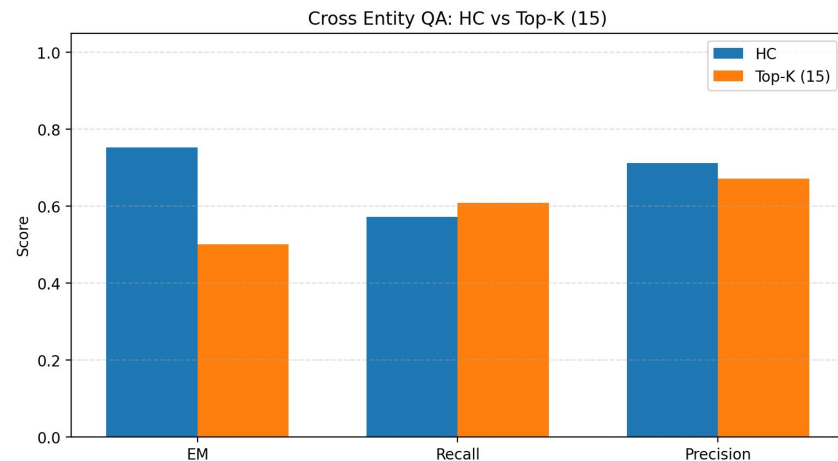
Musique: Multi-hop QA dataset, enforcing multi hop reasoning by combining single hop questions so each reasoning step relies on information from another

- HC increased Precision (**+12.73%**) wasn't translated into higher EM (**-19.25%**).
- Retrieving large amount of document significantly improved EM compared to Precision effect on EM



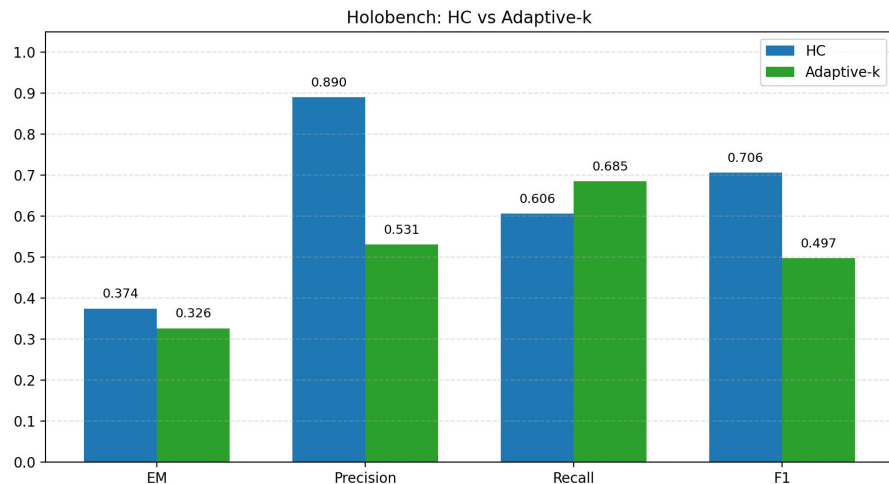
CrossEntityQA: Single-hop QA with large variability for number of relevant documents per query

- EM improved significantly when using HC (**+50%**)
- HC constantly achieved higher Precision and lower Recall



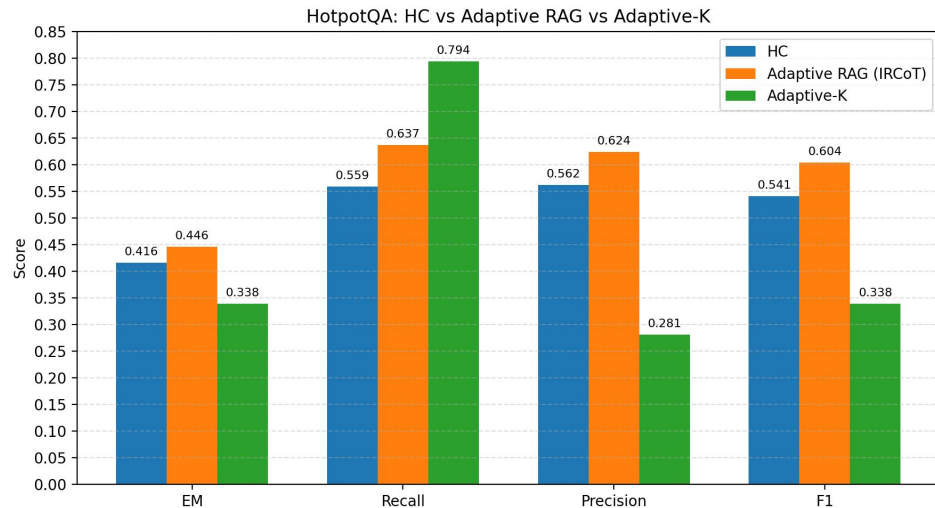
Results - Adaptive comparison

Holobench: Long context dataset, used for testing whether long-context language models can do “database-style” reasoning over text, combining many scattered facts.



- HC dominance over dynamic thresholding algorithm kept in Holobench dataset.
- Although HC achieved much higher Precision (**+67.6**) and F1 score (**+42.1**), its EM wasn't affected as much (**14.8%**)

HotpotQA: Multi-hop question answering dataset built from Wikipedia, designed to test whether a system can combine evidence from multiple articles.

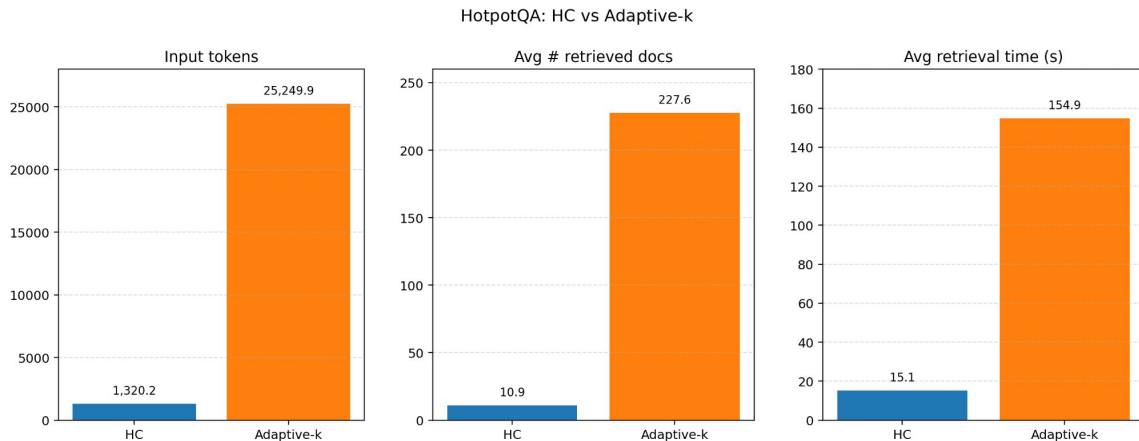


- HC achieved much higher precision than Adaptive-k (**90%** increase) which translates into higher EM (**23%** increase)
- HC couldn't beat Adaptive RAG on multi hop datasets when documents have dependance between them (**-6.7%** and **-10%** decreased EM and precision accordingly)

Results - Adaptive comparison

HotpotQA:

- HC was much faster (**+90.25**) and cheaper (**-94.77%** less input tokens) compared to other dynamic retrieval algorithm



Holobench:

- Efficiency performance was kept over different datasets as well.
- HC achieved better EM while also being much cheaper and faster to use!

